

Table 5.7 THEOPHYLLINE – summary of relative tolerance intervals bounds.

Reference value (µg/l)	0.05	0.10	0.50	1.00	2.50	10.00
Recovery (%)	117%	112%	104%	100%	101%	104%
β -ETI $\beta\% = 80\%$.	84%	94%	94%	88%	86%	96%
	150%	129%	114%	113%	116%	111%
β - γ -CTI with $\beta\% = 80\%$; $\gamma\% = 95\%$.	59%	81%	86%	78%	76%	91%
	176%	142%	121%	123%	125%	116%
Acceptance limit $\pm 25\%$	75%	75%	75%	75%	75%	75%
	125%	125%	125%	125%	125%	125%

Resource I β - γ Content tolerance interval (Excel).						
	A	B	C	D	E	
1	Resource I: β-γ-Content Tolerance Interval					
2	Assigned Reference Value (µg/L)	0.100				
3		Recovered concentration				
4		Replicate 1	Replicate 2	ni	SSi	
5	Series 1	0.114	0.113	2	5.00E-07	
6	Series 2	0.112	0.110	2	2.00E-06	
7	Series 3	0.104	0.101	2	4.50E-06	
8	Series 4	0.100	0.105	2	1.25E-05	
9	Series 5	0.113	0.107	2	1.80E-05	
10	Series 6	0.147	0.112	2	6.13E-04	
11	General parameters					
12	Number of series (I)	6	=COUNT(B5:B10)			
13	Number of measures (IJ)	12	=COUNT(B5:C10)			
14	Number of replicates (J)	2	=IF(D5*B12<>B13;"Error";D5)			
15	Residual sum of squares (SSW)	0.0006500	=SUM(E5:E10)			
16	Total sum of squares (SSt)	0.0016350	=DEVSQ(B5:C10)			
17	Inter-series sum of squares (SSB)	0.0009850	=B16-B15			
18	Repeatability variance (s^2_r)	0.0001083	=B15/(B13-B12)			
19	Temporary Between variance	0.0000443	=((B17/(B12-1))-B18)/B14			
20	Between-series variance (s^2_B)	0.0000443	=IF(B19<0;0;B19)			
21	Reproducibility variance (s^2_R)	0.0001527	=B20+B18			
22	Precision					
23	Recovered concentration	0.1115	=AVERAGE(B5:C10)			
24	Repeatability std. dev. (sr)	0.01041	=SQRT(B18)			
25	Between-series std. dev. (sB)	0.00666	=SQRT(B20)			
26	Intermediate Precision std. dev.. (sFI)	0.01236	=SQRT(B21)			
27	Trueness					
28	Relative bias (%)	11.5%	=(B23/B2)-1			
29	Recovery yield (%)	111.5%	=B23/B2			
30	beta-gamma-Content Tolerance Interval					
31	Tolerance (beta)	80%				
32	Confidence (gamma)	95%				
33	Repeatability mean square	0.00011	=B15/(B12*(B14-1))			
34	Between-series mean square	0.00020	=B17/(B12-1)			
35	Weighting coefficient (W)	9.29949	=(B12*(B33+(B14-1)*B34))/B34			
36	Normal law quantile	1.282	=NORMSINV((1+B31)/2)			
37	Coverage factor (kIC)	1.349	=B36*SQRT(1+(1/B35))			
38	Coefficient H1	3.365	=((B12-1)/CHIINV(B32;B12-1))-1			
39	Coefficient H2	2.669	=((B12*(B14-1))/CHIINV(B32;B12*(B14-1)))-1			
40	Weighing coefficient (W)	0.00036	=SQRT(((B34*B38/B14)^2)+((B33*B39*((B14-1)/B14))^2))			
41	CTI standard deviation (sIC)	0.02268	=SQRT((B33+MAX(0;(B34-B33)/B14)+B40))			
42	Lower bound content interval	0.081	=B23-B37*B41			
43	Upper bound content interval	0.142	=B23+B37*B41			

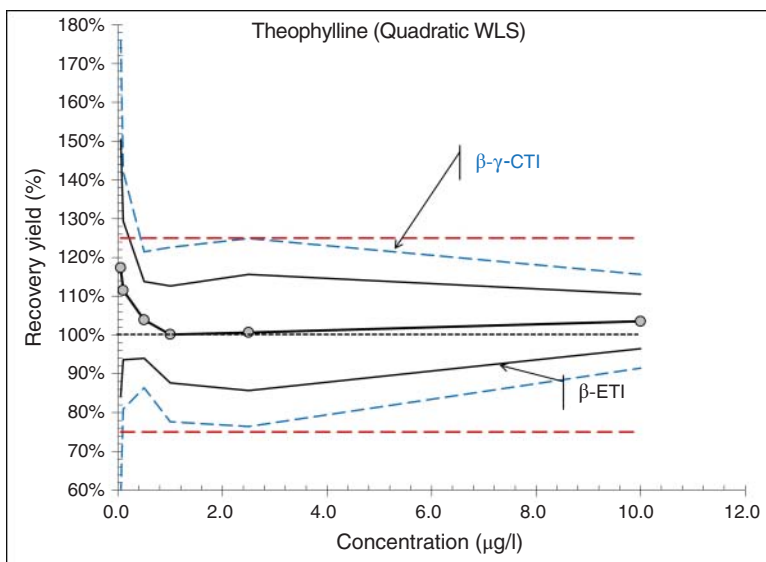


Figure 5.10 THEOPHYLLINE – accuracy profile with the two tolerance intervals ($\beta\% = 80\%$, $\gamma\% = 95\%$). Inverse-predicted concentrations are obtained with WLS quadratic models.

In Resource I worksheet, the calculation of the standard deviation of the interval, denoted s_{IC} , is a little bit complicated. It requires two weighting coefficients, H_1 and H_2 , which contain the quantiles of the χ^2 law that are calculated as follows:

Coefficient	Degrees of freedom	Excel formulas	Value
H_1	$I - 1 = 6 - 1 = 5$	<code>=CHIINV(0.95;5)</code>	3.36
H_2	$I \times (J - 1) = 6 \times 1 = 6$	<code>=CHIINV(0.95;6)</code>	2.67

Up to row 30, all formulas are the same for both TIs. Therefore, it is possible to combine them into one single worksheet. In rows 31 and 32, the two probability values associated with this type of interval are entered. Some formulas are quite long and made visible in column C. The worksheet could be improved by reorganizing Eqs. (5.14)–(5.17). It is also possible to put together in a single worksheet the three programs Resource E, Resource H, and Resource I. To present the details of the calculation and facilitate the adaptation and verification by any laboratory, they are kept separate. In Section 5.4.4, the influence of the choice of probabilities β and γ is discussed in more detail, especially in view of the various official validation guides.

5.4 Accuracy Profile: Special Topics

Besides its application for method validation, MAP can also be of some help in optimizing some final steps of the operating procedure or checking if the planning of the validation study was correct or may be improved.

Table 5.8 THEOPHYLLINE – coefficients of all quadratic models adjusted to the series of calibration data.

Series	Algorithm	a_0	a_2	a_3	r^2 (%)	AIC
1	OLS	1.0403	11.779	0.0837	99.5	59.50
1	WLS	0.0489	16.312	-0.4281	95.5	39.46
2	OLS	0.5482	15.258	-0.1924	100.0	23.87
2	WLS	0.0375	17.828	-0.485	99.4	20.94
3	OLS	0.1614	17.506	-0.4985	99.9	39.04
3	WLS	0.1866	17.697	-0.5232	99.0	25.05
4	OLS	0.3039	6.5998	-0.0847	99.9	30.69
4	WLS	0.1408	7.2525	-0.1574	99.4	3.67
5	OLS	0.1874	8.5221	-0.118	100.0	-1.09
5	WLS	0.0391	9.2555	-0.2014	99.7	0.86
6	OLS	0.3315	6.4809	-0.0855	100.0	11.69
6	WLS	0.0398	7.7705	-0.2306	99.3	5.56

5.4.1 Choose the Best Calibration Model

In Section 2.1, it was mentioned that various calibration models can be adjusted to the same dataset, using either OLS or WLS estimators. The first question was to select the best calibration model. In statistical literature, different parameters are proposed, such as the coefficient of determination r^2 or the Akaike information coefficient (AIC). Regarding the example in Table 2.2, no decision could be made because these parameters are too close. In other words, they are not sensitive enough. Table 5.8 summarizes the coefficients for all quadratic models separately adjusted to the six-calibration series of the THEOPHYLLINE dataset. These values were obtained using the short Python script of Resource B applied to the five other series of calibration data (not provided).

The situation seems even more confusing as the calibration model coefficients are highly variable. It is even surprising to have consistent inverse predicted concentrations when the coefficients seem to change so much. This confirms two basic proposals:

- Validation study should be organized into several series of measurements to verify that the quantification capacity of the method is not affected over time by various sources of variation.
- Calibration model is a critical issue but not the key issue. Using calibration parameters, such as sensitivity or linearity, is not pertinent to assess method validated range.

It is possible to construct several MAPs with different inverse-predicted concentrations obtained either by OLS or WLS calibration models. When differences

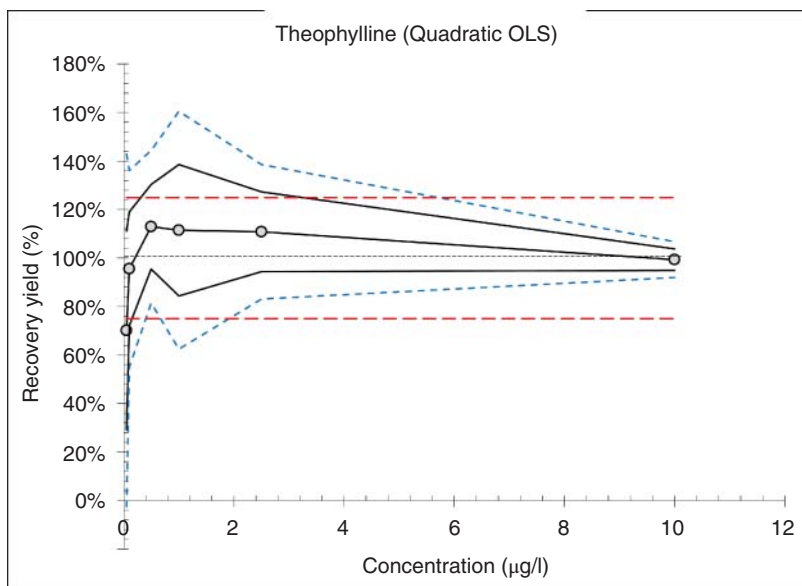


Figure 5.11 THEOPHYLLINE – accuracy profile (MAP) with the two tolerance intervals obtained for OLS quadratic models ($\beta\% = 80\%$, $\gamma\% = 95\%$).

appear among these profiles, it may help to make the final decision about the best calibration model. The profile based on OLS models is illustrated in Figure 5.11 and must be compared to Figure 5.5 obtained with WLS models. The decision is evident: the new MAP is much less favorable than the previous one as the validated range is clearly diminished; the OLS model is unsatisfactory. Moreover, a proposal was to define the *LOQ* as the lowest limit of the validated range. It was $0.129 \mu\text{g/l}$ with WLS curves, while it is at $2.698 \mu\text{g/l}$ with the new MAP. This example illustrates the possible use of MAP as an ultimate optimization tool. For future routine application of the theophylline method of analysis, for each series, the calibration curve *must* be a second-order polynomial calculated by weighted least-squares regression.

In Section 2.3.2 about WLS, it was underlined that the choice of the weighting may have some consequences on the performance of the inverse prediction. All presented computations were done using the questionable $1/X^2$ weighting factor. Even better results may be obtained with another more adapted choice.

5.4.2 Apply Consistent Experimental Design

Since the 1930s, many published books focused on the theory of experimental design. The early contribution of R.A. Fisher (1890–1962), called *The design of experiments* is the keystone of the theory. In the introduction, it is explained why, in any experimental work, it is fundamental to organize the trials according to well-established rules:

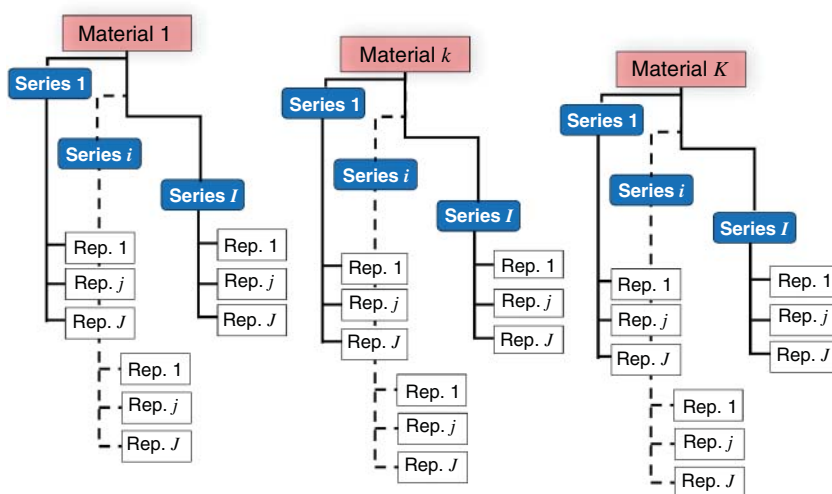


Figure 5.12 Schematic representation of the experimental design to be used to build a relevant accuracy profile. Material means reference material of known concentration.

...a criticism frequently levelled at test results is that the experiment was poorly designed and, therefore, poorly conducted. Assuming that the experimenter did what he intended to do, this point comes down to the question of the design and logical structure of the experiment.

Since this work, there has been much evidence that the relevance of experimental data fundamentally depends on the conditions it was collected. A consequent number of publications propose optimal experimental designs adapted to different contexts [29]. Therefore, to obtain a relevant accuracy profile, it is logical also to use a relevant experimental design based on perquisite rules. Figure 5.12 exemplifies these rules.

Four rules are proposed to design a consistent validation experiment.

Rule 1. Select at least three validation materials ($K \geq 3$) with known (or known with a defined uncertainty) concentration levels covering the validation range. They can combine different matrices as far as they have comparable effects. The number of three is minima as it is used to verify the trueness is linear, i.e. the proportionality exists between the known contents of reference material and the recovered contents after application of the whole analytical procedure. A greater number of validation materials would be preferable, as only three may lead to disappointment.

Rule 2. Perform under intermediate precision conditions at least three series of measurements ($I \geq 3$) for each validation material, i.e. so that several sources of uncertainty, as representative as possible of the routine condition, come into play from one series to another.

Rule 3. Provide a minimum of two replicates per series ($J \geq 2$) on the same validation material, performed under repeatability conditions. A number of official guides have a higher requirement ($J = 3$ to 6).

Rule 4 (optional but allows for simplified calculations later). Ensure all series have the same number of replicates to have a balanced experimental design.
